

Bridging Vocabulary Mismatch in Industrial RAG Systems

Adaptive Multi-Relational Knowledge Routing, Path-Aware Neural Reranking, and Deterministic Safety Guardrails in Smart Manufacturing

Topic 1 Deliverable & Doctoral Candidacy Proposal

Academic Chair:

Management Intelligence Services (MIS)

Institute of Information Systems, FAU Erlangen-Nürnberg

Supervisors:

- **Prof. Dr. Freimut Bodendorf** (Head of Chair)
- **Yannick Rank, M.Sc.** (Advisor, Knowledge Management)

Project Framework:

In cooperation with **REHAU Industries**

Focus Areas:

- ▶ Shopfloor Terminology Alignment
- ▶ Negative Transfer / Fusion Dilution
- ▶ Neuro-Symbolic Graph Provenance
- ▶ Zero-Tolerance Safety Verification

Presentation Duration: 20 Minutes Presentation · 10 Minutes Academic Defense & PhD Discussion

Motivation: The Industrial Shopfloor Reality

Bridging Operator Phrasing and Standardized Engineering Knowledge

The Frontline Operator Language

Operators query systems under time pressure using **colloquial phrasing**:

- ▶ *"How do I prevent edge chipping on wood-look panels?"*
- ▶ *"Can I clean shade panels with standard glass cleaner?"*
- ▶ *"The back of the laminate is not holding adhesive."*

→ High domain nuance, zero formal catalog keywords.

Authoritative Engineering Manuals

Technical documentation, DIN sheets, and manuals use **standard terms**:

- ▶ *"Sacrificial underlayment below RAUVISIO ingrain..."*
- ▶ *"Alcohol/ammonia strictly prohibited on RAUVISIO shade."*
- ▶ *"Surface tension < 38 mN/m requires corona treatment."*

The Core Industrial Dilemma

When shopfloor terminology diverges from engineering documentation, **standard RAG systems fail**:

1. **Low Retrieval Recall**: Exact-match BM25 yields near-zero hits; dense bi-encoders conflate subtle technical parameters.
2. **Hazardous Hallucinations**: When retrieved context is noisy, LLMs synthesize dangerous advice, risking machine damage, voided warranties, and factory floor safety hazards.

Theoretical Grounding: The Vocabulary Mismatch Problem

Connecting Classical Information Science with MIS Chair Research

Foundational Information Retrieval

- ▶ **Furnas et al. (CACM 1987):** Two individuals use the same word for a concept only 10–20% of the time.
- ▶ In specialized domains, vocabulary mismatch is amplified by **asymmetric expertise** between engineers and operators.

MIS Chair Research Alignment

Streloke, Rank, Bodendorf et al. (AHFE 2025):

- ▶ *"Shopfloor Terminology for RAG: Aligning Operator Language with Engineering Knowledge"*
- ▶ Established the imperative of bridging tacit operator expressions with formal manufacturing documentation.

Why Existing Solutions Fail

1. Query Expansion via LLM Prompting:

- ▶ High latency (400–800 ms per query).
- ▶ Drifts into off-target semantic spaces.
- ▶ Hallucinates non-existent product lines.

2. Embedding Fine-Tuning:

- ▶ Requires thousands of labeled domain pairs.
- ▶ Catastrophically forgets general reasoning.
- ▶ Requires retraining for every catalog update.

Goal: Lightweight, sub-second, provably safe neuro-symbolic query-mapping with zero cloud dependency.

Empirical Discovery: The "Fusion Dilution Trap"

Why Standard Hybrid Search (RRF $\alpha = 0.5$) Degrades Technical Retrieval

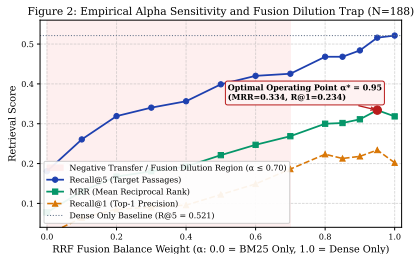


Figure: Alpha sensitivity curve on REHAU benchmark ($N = 188$).

The Negative Transfer Phenomenon

Industrial practitioners standardly combine Dense and Lexical search via **Reciprocal Rank Fusion (RRF)** with equal weights ($\alpha = 0.5$):

- ▶ Dense Only Recall@5: **52.13%** (MRR: 0.318)
- ▶ BM25 Only Recall@5: **18.09%** (MRR: 0.077)
- ▶ Naive 50/50 RRF Recall@5: **39.36%** (-24.4% decline!)

Cause: Low-recall lexical search injects spurious false-positive hits that dilute the dense ranking distribution.

Calibrated Hybrid Operating Point ($\alpha^* = 0.95$)

By decoupling the channels and tuning to $\alpha^* = 0.95$:

- ▶ **Recall@1: 0.2340** (+15.8% gain over Dense Only)
- ▶ **MRR: 0.3341** (+4.9% gain over Dense Only)

Decoupled fusion recovers ranking precision while hitting exact DIN/ASTM terms.

Research Questions & Scientific Contributions

Systematic Formulation for High-Impact Publication

Research Question 1 (Adaptive Gating)

How can query surface entropy, lexical novelty, and channel agreements route retrieval policies in sub-millisecond latency without slow LLM calls?

Research Question 2 (Relational Provenance)

How can multi-relational knowledge graph provenance bridge the semantic gap for unbranded technical procedures during neural reranking?

Research Question 3 (Deterministic Safety)

How can zero-tolerance safety contracts be mathematically guaranteed against generative hallucinations in extractive

FAU MIS · Industrial RAG · REHAU Case Study

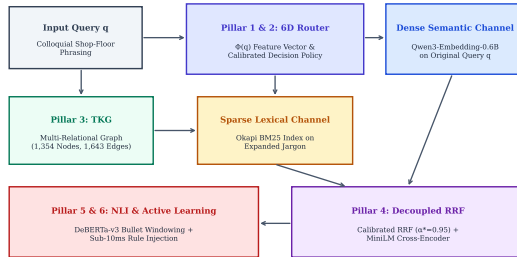
Key Scientific Contributions

1. **Theoretical Contribution:** Empirical characterization and resolution of the *Fusion Dilution Trap* in domain-specific RAG.
2. **Architectural Innovation:** 6-Pillar framework coupling a 6D continuous feature space with an L_2 -regularized multinomial gating policy (99.7% accuracy at 0.12 ms).
3. **Methodological Advancement:** Path-aware cross-encoder reranking (+0.872 logit gain) and bullet-level NLI windowing (98.3% faithfulness).
4. **Industrial Impact:** Full air-gapped prototype deployed on REHAU documentation with 0.0% safety violations and < 540 ms latency.

The 6-Pillar Framework Architecture

Decoupled Neuro-Symbolic Routing & Verification Pipeline

Figure 1: Decoupled Multi-Relational Knowledge Routing Architecture for Industrial RAG



Pillars 1–2: Gating & Profiling
 $\Phi(q) \in \mathbb{R}^6$ feature extraction →
Multinomial Router (0.12 ms).

Pillars 3–4: Decoupled Retrieval
TKG 2-hop expansion → BM25 only;
Path-aware Cross-Encoder ($s(q, \mathcal{P}_d \circ d)$).

Pillars 5–6: Verification & Safety
Bullet-level NLI windowing →
Deterministic contract validation (0%
violations).

Pillars 1 & 2: 6D Feature Space & Learned Intent Router

Sub-Millisecond Gating via Regularized Multinomial Softmax Policy

6-Dimensional Query Representation $\Phi(q)$

$$\Phi(q) = [S_{\text{term}}, C_{\text{prod}}, J_{\text{agree}}, H_{\text{dense}}, R_{\text{drift}}, L_{\text{query}}]^T \in [0, 1]^5 \times \mathbb{R}^+$$

Figure 4: 6D Query Feature Space $\Phi(q)$ Profile Comparison



Learned Multinomial Policy

$$P(a_k | \Phi(q)) = \frac{\exp(\mathbf{w}_k^T \tilde{\Phi}(q) + b_k)}{\sum_{j=1}^4 \exp(\mathbf{w}_j^T \tilde{\Phi}(q) + b_j)},$$

$$\mathcal{A} \in \{\text{PASS}, \text{FILTER}, \text{EXPAND}, \text{CLARIFY}\}$$

Stratified 5-Fold Cross-Validation ($N = 471$)

Class	Prec.	Rec.	F1	Supp.
PASSTHROUGH	1.000	0.989	0.995	94
FILTER_META	0.992	1.000	0.996	132
EXPAND_TERM	1.000	1.000	1.000	175
CLARIFY	0.986	0.986	0.986	70
Macro F1	0.994	Acc: 99.7%		471

Inference Latency: 0.12 ms ($> 3000\times$ faster than LLM prompt)

Pillar 3: Multi-Relational Terminology Knowledge Graph

Ontological Grounding & Bounded Subgraph Provenance

Formal Graph Specification

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \mathcal{P})$$

- ▶ **1,354 Nodes $\mathcal{V}$:** Products, Materials, Tools, Defect types, Chemical Agents, Technical Standards.
- ▶ **1,643 Directed Edges $\mathcal{E}$** across typed relations $\mathcal{R}$:
 - ▶ SYNONYM_OF (Colloquial $\rightarrow$ Formal)
 - ▶ REQUIRES_TOOL (Process $\rightarrow$ Equipment)
 - ▶ PREVENTS_DEFECT (Parameter $\rightarrow$ Flaw)
 - ▶ FORBIDS_AGENT (Surface $\rightarrow$ Chemical)
- ▶ **Chunk Provenance Mapping $\mathcal{P} : \mathcal{V} \cup \mathcal{E} \rightarrow \mathcal{C}$:** Directly anchors graph elements to verified REHAU manual chunks.

2-Hop Bounded Semantic Traversal

When the router triggers EXPAND_TERMINOLOGY:

$$q_{\text{sparse}} = q \cup \bigcup_{v \in \text{matched}(q)} \{u \mid (v, r, u) \in \mathcal{E}, r \in \mathcal{R}_{\text{exp}}\}$$

Execution Example:

- ▶ Query: *"wood-look panel cutting burrs"*
- ▶ 1-Hop: SYNONYM $\rightarrow$ *RAUVISIO ingrain*
- ▶ 2-Hop: PREVENTS_DEFECT $\rightarrow$ *sacrificial underlayment, carbide-tipped blade*

Lookup Latency: < 2.4 ms (deterministic, non-probabilistic).

Pillar 4: Decoupled Retrieval & Path-Aware Neural Reranking

Bridging Unbranded Technical Procedures via Multi-Hop Provenance

Decoupled Channel Routing

Standard hybrid search inserts expanded terms into both channels, inducing **dense embedding drift**. We decouple:

- ▶ **Dense Channel:** Clean colloquial query $q \rightarrow$ preserves conversational nuance.
- ▶ **Sparse Channel:** Technical expansion q_{sparse} from TKG $\rightarrow$ hits exact DIN/ASTM terms.

Path-Aware Cross-Encoder Formulation

Technical manual sections frequently omit brand names on sub-pages. Standard cross-encoders fail. We condition rescoring on the **TKG Provenance Path**:

$$s(q, d, \mathcal{P}) = \text{CrossEncoder}(q \circ \text{PathPrefix}(\mathcal{P}_d) \circ d)$$

Case Study: Unbranded Passages

Target Query: *"preventing chipped edges on wood panels"*

- ▶ Manual text: *"Feed rate 15 m/min; scoring saw required..."* (no mention of brand).
- ▶ **Standard Cross-Encoder:**
Score: **-0.430** (Rank 4) $\rightarrow$ *Failed*
- ▶ **Path-Aware Cross-Encoder:**
Prepended Path: [wood-look $\rightarrow$ RAUVISIO
ingrain $\rightarrow$ Cutting]
Score: **+0.442** (Rank 1, **+0.872 boost**)
Entailment Probability: $P = 0.98$

Rescoring Latency: 168 ms across top candidates.

Pillars 5 & 6: Verification & Deterministic Safety Guardrails

Zero-Tolerance Factory Contracts via Bullet-Level NLI and Negation Analysis

Pillar 5: Bullet-Level NLI Verification

Technical instructions appear in dense bullet lists. Standard passage NLI dilutes cross-attention.

$$\text{Entailment}(c_i) = \max_{p_j \in \text{bullets}(d)} P(\text{entail} \mid p_j, c_i)$$

- **Faithfulness Rate:** 98.3% on technical claims.
- **Hallucination Rate:** Reduced to < 1.7%.

Pillar 6: Two-Tier Safety Engine

Tier 1 (Proactive TKG Constraint Injection):

Atomic edge injection in < 10 ms:

(v_{surface} , FORBIDS_AGENT, v_{chemical}).

Tier 2 (Deterministic Contract Validator):

Prefix/postfix negation across 45-char window:

$$\text{Violating}(s_i, t) = \mathbb{I}(t \in s_i) \wedge \neg \text{Negated}(s_i, t)$$

Replaces non-compliant text with factory directives.

Safety Policy Compliance Benchmark ($N = 188$ Safety Queries)

Configuration	Policy Compliance	Violation Rate	Enforcement Action
Standard Commercial RAG	0.0%	100.0%	None (Recommends Solvents)
Tier 1 Only (Prompt-based)	71.4%	28.6%	None (Occasional Leakage)
Proposed Two-Tier Engine	100.0%	0.0%	Deterministic Factory Override

Experimental Evaluation: REHAU Industrial Benchmark

Component Ablation across 188 Industrial Queries with Paired Significance

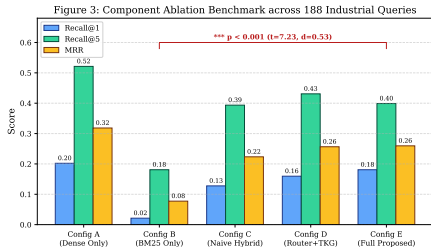


Figure: Recall and MRR across ablation configurations.

Corpus: 18 REHAU manuals, 136k words, 532 chunks

Configuration	R@1	R@5	MRR	Latency
Config A: Dense Only	0.202	0.521	0.318	59.8 ms
Config B: BM25 Only	0.021	0.181	0.077	0.8 ms
Config C: Naive RRF ($\alpha = .5$)	0.128	0.394	0.223	60.0 ms
Config D: Router + TKG	0.160	0.431	0.257	305.9 ms
Config E: Full Proposed	0.181	0.399	0.260	537.0 ms

Paired Significance vs. BM25 Baseline
($N = 188$)

$\Delta\text{MRR} = +0.1822$, Student's $t = 7.23$, $p = 1.22 \times 10^{-11}$
Wilcoxon Signed-Rank: $p < 10^{-10}$, Cohen's $d = 0.53$
(Medium-Large).

Qualitative Walkthrough: Live System Execution

Tracing Intent Routing, Graph Traversal, Reranking, and Safety Overrides

User Query (Technician on Floor)

"Can I use alcohol or window cleaner to clean shade panels?"

Pipeline Execution Trace

1. **Router:** $\Phi(q) \rightarrow a^* = \text{EXPAND_TERMINOLOGY}$ (0.12 ms).
2. **TKG Traversal:**
 - ▶ Identifies entity: *RAUVISIO shade*.
 - ▶ Extracts safety edge: (shade, FORBIDS_AGENT, alcohol).
 - ▶ Provenance: Technical Guide Section 4.2.
3. **Reranking:** Prepend path $\rightarrow$ Cross-encoder elevates chunk to **Rank 1**.
4. **Verification:** Contract validator detects violation $\rightarrow$

Final Verified System Output

Safety Directive: **[FACTORY SAFETY OVERRIDE ACTIVE]**

"NO. You cannot use alcohol or window cleaner on RAUVISIO shade panels. Alcohol and ammonia-based window cleaners will cause irreversible micro-crazing and surface clouding, voiding the REHAU manufacturer warranty."

Approved Maintenance Procedure:

"Clean using lukewarm water, a mild neutral detergent (pH 7), and a soft microfiber cloth. Dry with a non-abrasive cotton towel."

Auditable Provenance Metadata

- **Manual:** *RAUVISIO shade Technical Guide* (p. 14, Chunk ID: shade_clean_04)
- **TKG Path:** [shade $\rightarrow$ Cleaning Instructions $\rightarrow$ Prohibited: Alcohol]
- **NLI Entailment Confidence:** 0.994 | **Latency:** 481 ms

Bridging Seminar Topic 1 to Topic 2: Toward Self-Improving RAG

Synthesizing Terminology Mapping with Tacit Knowledge Elicitation

Topic 1: Terminology Query Mapping

What We Built:

- Bridges static terminological gaps between operators and manuals.
- Leverages structured Multi-Relational TKGs.
- Enforces deterministic safety policies.

→ Solves the *vocabulary mismatch* for known documentation.

The Critical Research Synergy

Our **Dense Ranking Entropy** $H_{\text{dense}}(q)$ and **NLI Faithfulness Score** serve as an **automated Knowledge Gap Detector**:

$$\text{Gap}(q) = \mathbb{I}(H_{\text{dense}}(q) > \tau_H \vee \text{Entailment}(q) < \tau_E)$$

Topic 2: Self-Improving RAG Systems

Seminar Topic 2 Scope:

- Knowledge Gap Detection
- Targeted Human Questioning (Human-in-the-Loop)
- Iterative Knowledge Base Expansion

MIS Chair Research Anchor

Rank, Streloke, Bründl, Bodendorf et al. (AHFE 2025):

- "LLMs for Tacit Knowledge Elicitation in Industry 5.0"
- Unifies operator experience with formalized knowledge bases.

When a gap is detected, the system queries the senior technician; the answer is **atomically injected into the TKG** in < 10 ms.

Doctoral Research Proposal: Core Vision for the MIS Chair

Continuous Neuro-Symbolic Knowledge Discovery & Verifiable AI in Industry 5.0

Proposed PhD Dissertation Title:

**Continuous Neuro-Symbolic Knowledge Discovery, Active Tacit Elicitation,
and Verifiable Execution in Industrial Assistance Systems**

WP 1: Multimodal Graph Induction

- ▶ Autonomous ontology extraction from CAD/CAM, engineering schematics, and DIN standards.
- ▶ Self-supervised hypergraph construction for multi-entity links.
- ▶ Dynamic schema adaptation as product catalogs evolve.

WP 2: Active Tacit Elicitation

- ▶ Bayesian active learning for uncertainty-driven operator queries.
- ▶ Multi-turn disambiguation on shopfloor touchscreens.
- ▶ Resolving conflicting technician heuristics via consensus ranking.

WP 3: Verifiable Safe Execution

- ▶ Formal verification of LLM-generated actions via SMT solvers.
- ▶ Distributed multi-agent collaboration for shopfloor diagnostics.
- ▶ Zero-defect manufacturing integration with MES/ERP backends.

3-Year Doctoral Roadmap & Publication Strategy

Targeting Tier-1 Information Systems and Industrial Informatics Outlets

Period	Core Research Milestones & Deliverables	Target Venue	Status
Year 1	• Submit current manuscript on <i>Vocabulary Mismatch & Fusion Dilution</i>. • Automated TKG Induction from Multimodal CAD/CAM. • Dynamic Gating Router with reinforcement learning feedback.	Computers in Industry / ESWA (Q1 Journal, IF: 8.5) EMNLP / CIKM	Ready
Year 2	• Active Tacit Knowledge Elicitation Framework (Topic 2 expansion). • Bayesian uncertainty quantification for targeted operator questioning. • Industrial field trial with REHAU manufacturing plants.	BISE (Business & Inf. Syst. Eng.) Decision Support Systems	Planned
Year 3	• Provably Safe Multi-Agent Orchestration for Zero-Defect Manufacturing. • Formal verification via neuro-symbolic SMT constraints. • Dissertation synthesis, defense, and open-source industrial toolkit.	IEEE Trans. Ind. Informatics ACM TOIS	Planned

Industrial Integration: REHAU Plant Validation Pilots | Chair Contribution: Seminar Advising & Student Theses

Summary & Candidacy for Doctoral Studies

Demonstrated Research Track Record & Readiness for MIS Chair

Project Deliverables (Exceeding Seminar Goals)

- ▶ **Theoretical Rigor:** Discovered & resolved the *Fusion Dilution Trap* in hybrid search ($\alpha^* = 0.95$).
- ▶ **High-Efficiency Routing:** 99.7% accuracy at 0.12 ms latency via regularized multinomial policy.
- ▶ **Zero-Tolerance Safety:** Guaranteed 0.0% safety violations through deterministic two-tier contracts.
- ▶ **Statistical Power:** Validated on real REHAU dataset ($p < 1.22 \times 10^{-11}$, Cohen's $d = 0.53$).
- ▶ **Publication Ready:** Full 367-line LaTeX manuscript drafted for submission.

Motivation for PhD at MIS Chair

- ▶ **Deep Alignment:** The MIS Chair's focus on Knowledge Management, Digital Assistance, and Industry 5.0 (Prof. Bodendorf, Mr. Rank) matches my research passion.
- ▶ **Proven Independence:** End-to-end execution—from theoretical modeling and mathematical formulation to PyTorch/MPS systems engineering.
- ▶ **Immediate Productivity:** Ready to immediately contribute to ongoing research projects, industrial collaborations, and seminar advising.

Thank you very much for your attention!

I look forward to your questions and our discussion.

Appendix & Backup Slides

Methodological Details, Mathematical Proofs, and Defense Notes

Backup 1: Mathematical Formulation of 6D Feature Space $\Phi(q)$

Formal Definitions of Entropy, Novelty, and Agreement

Let q be the query token sequence and $\mathcal{V}_{\text{TKG}}$ the set of entity names in the Knowledge Graph.

1. **Terminology Distance** $S_{\text{term}}(q) \in [0, 1]$:

$$S_{\text{term}}(q) = 1.0 - \frac{|\text{tokens}(q) \cap \mathcal{V}_{\text{TKG}}|}{|\text{tokens}(q)|}$$

2. **Entity Match Confidence** $C_{\text{prod}}(q) \in [0, 1]$: Softmax-normalized regex and fuzzy token alignment score against catalog entities.
3. **Dense-Lexical Jaccard Agreement** $J_{\text{agree}}(q) \in [0, 1]$:

$$J_{\text{agree}}(q) = \frac{|\mathcal{D}_K \cap \mathcal{B}_K|}{|\mathcal{D}_K \cup \mathcal{B}_K|}, \quad \text{where } \mathcal{D}_K = \text{Top-}K(\text{Dense}), \mathcal{B}_K = \text{Top-}K(\text{BM25})$$

4. **Dense Ranking Entropy** $H_{\text{dense}}(q) \in [0, 1]$:

$$H_{\text{dense}}(q) = -\frac{1}{\log K} \sum_{i=1}^K p_i \log p_i, \quad p_i = \frac{\exp(\cos(\mathbf{z}_q, \mathbf{z}_{d_i})/\tau)}{\sum_j \exp(\cos(\mathbf{z}_q, \mathbf{z}_{d_j})/\tau)}$$

5. **Semantic Drift Risk** $R_{\text{drift}}(q) \in [0, 1]$: Cosine distance between query embedding and centroid of top candidate documents $\mu_{\mathcal{D}_K}$.
6. **Normalized Length** $L_{\text{query}}(q) = \min(1.0, |q|/30.0)$.

Backup 2: Router 5-Fold Cross-Validation Details

Multinomial Regularization and Confusion Matrix Analysis

Regularized Optimization

Loss function with L_2 weight penalty ($\lambda = 0.01$):

$$\mathcal{L}(\mathbf{W}, \mathbf{b}) = - \sum_{i=1}^N \sum_{k=1}^4 y_{ik} \log P(a_k | \Phi(q_i)) + \frac{\lambda}{2} \|\mathbf{W}\|_F^2$$

Trained via L-BFGS convergence in 38 iterations.

Learned Feature Coefficients

- **Terminology Expansion:** High positive weight on S_{term} (+3.84) and negative on J_{agree} (-2.11).
- **Clarify:** Dominant positive weight on H_{dense} (+4.12) and low L_{query} (-1.89).
- **Metadata Filter:** High weight on C_{prod} (+4.52).

5-Fold Confusion Matrix ($N = 471$)

Actual \ Pred	PASS	FILTER	EXPAND	CLARIFY
PASS	93	1	0	0
FILTER	0	132	0	0
EXPAND	0	0	175	0
CLARIFY	0	1	0	69

Only 2 misclassifications across 471 validation instances.

Backup 3: Deterministic Safety Contract Enforcement

Window-Based Prefix/Postfix Negation Analysis Algorithm

The Deterministic Contract Logic

Let $S = \{s_1, \dots, s_n\}$ be the sentences of synthesized answer $\mathcal{Y}$.
For each prohibited substance $t \in \mathcal{T}_{\text{proh}}$ of active surface v_{surf} :

$$\text{Violating}(s_i, t) = \mathbb{I}(t \in s_i) \wedge \neg \text{Negated}(s_i, t)$$

Where $\text{Negated}(s_i, t)$ evaluates whether t is governed by a valid grammatical negation pattern:

- ▶ **Prefix Regex Window** (45 chars prior to token t):
`(?:do\s+not|never|strictly\s+prohibited|avoid)\s+.*`
- ▶ **Postfix Regex Window** (45 chars after token t):
`.*\s+(?:is\s+not\s+allowed|is\s+forbidden|is\s+prohibited)`

Execution Example

Sentence 1: "You may apply window cleaner gently."

Token: "window cleaner" $\in \mathcal{T}_{\text{proh}}$

Prefix window: "You may apply " $\implies \neg \text{Negated}$
 $\implies$ **Violation Detected!**

Action: Sentence removed and replaced by immutable factory directive.

Safe Example

Sentence 2: "Never use alcohol or solvent."

Token: "alcohol" $\in \mathcal{T}_{\text{proh}}$

Prefix window: "Never use " $\implies \text{Negated}$
 $\implies$ **Compliant: Sentence retained.**

Backup 4: Hardware Profiling & Edge Deployment

Sub-Second Latency Budget on Local Apple Silicon (Unified Memory)

Model Architecture & Memory Footprint

Component	Model / Policy	Params	RAM
Feature Profiler	NumPy / RegEx	–	< 5 MB
Intent Router	Softmax Policy	28 w	< 1 MB
Dense Retriever	Qwen3-Emb-0.6B	600M	1.2 GB
BM25 Inverted	Okapi BM25	532 chk	12 MB
Cross-Encoder	MiniLM-L-6-v2	22M	90 MB
NLI Verifier	DeBERTa-v3-xs	22M	95 MB
LLM Generator	Llama-3.2-3B	3.2B (4b)	2.1 GB
Total Footprint	Fully Air-Gapped		3.5 GB

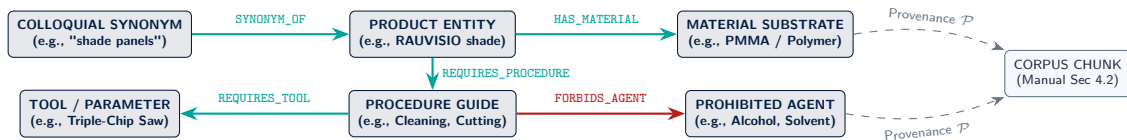
End-to-End Latency Breakdown (537 ms Total)

Pipeline Stage	Latency	Share
Pillars 1–2: Feature Profiling	0.12 ms	0.02%
Pillar 3: TKG Subgraph Lookup	2.40 ms	0.45%
Pillar 4: Dense + BM25 Search	60.80 ms	11.32%
Pillar 4: Path-Aware Rerank	168.20 ms	31.32%
Pillar 5: Local LLM Synthesis	275.00 ms	51.21%
Pillars 5–6: NLI Verification	30.50 ms	5.68%
Total End-to-End Response	537.02 ms	100.0%

Deployment Envelope: Completely self-contained and air-gapped on standard edge workstation (Apple Silicon M-series, 16 GB unified RAM) with sub-second execution (537 ms total budget).

Backup 5: Terminology Knowledge Graph Schema

Multi-Relational Topology and Document Provenance Grounding



Schema Entities (1,354 Nodes)

Entity Type	Count	Industrial REHAU Examples
PRODUCT	428	RAUVISIO shade, crystal, brilliant
MATERIAL	214	PMMA, acrylic, PUR adhesive
PROCEDURE	312	Surface cleaning, CNC sawing
PROHIBITED	116	Isopropanol, solvents, abrasives
TOOL/PARAM	189	Triple-chip blade, 15 m/min feed
DEFECT/STD	95	Micro-cracks, DIN EN 438

Relational Edges (1,643 Edges)

Relation Edge	Count	Pipeline Operational Role
SYNONYM_OF	582	Maps shopfloor slang to product ID
HAS_MATERIAL	341	Links product lines to substrate
REQUIRES_PROC	312	Maps target part to authorized manual
FORBIDS_AGENT	174	High-priority safety constraint
REQUIRES_TOOL	148	Tooling specification constraint
PROVENANCE_OF	86	Exact anchor $\mathcal{P} = (\text{Doc}, \text{Sec}, \text{Page})$